

SYNTHETIC DEMONSTRATION

OCVS target sizing

Recommended target: 5 BM.Standard.E5.192-192 hosts

The design covers all 248 workload VMs and remains within CPU and configured-memory limits after any one host becomes unavailable.

WORKLOAD VMS

248

17 powered off

CONFIGURED VCPU

2,216

all workload VMs

CONFIGURED RAM

6.90 TiB

binding constraint

VCF LICENSE CORES

960

full silicon

Design input

INPUT	VALUE
Sizing policy	Oracle default sizing policy
Workload scope	All non-template VMs, including powered-off VMs
CPU planning ratio	4 configured vCPU per configured physical core
Planning headroom	20% CPU and configured memory; 25% storage
Resilience	Capacity checked with one host unavailable
Topology	Consolidated target cluster

Why five hosts

Five is not a three-host minimum plus a blindly added spare. It is the largest calculated floor: the configured-memory requirement with one host unavailable. With five purchased hosts, four remain usable during maintenance or a host failure.

Decision signal

Configured memory, not CPU, determines this design. The result should be validated with observed utilization, reservations, NUMA requirements, and application constraints before procurement.

DETERMINISTIC SIZING

How the host count is set

The engine calculates independent capacity floors, applies provider constraints, and chooses the largest result. Every number can be traced back to inventory, policy, or target-node specifications.

Constraint floors

CONSTRAINT	HOSTS	RATIONALE
Normal CPU capacity	4	2,216 vCPU at 4:1 with 20% headroom
Normal configured memory	4	6.90 TiB with 20% headroom
One-host-loss CPU	4	Remaining hosts stay inside the CPU envelope
One-host-loss memory	5	Four remaining hosts carry planned memory
OCVS provider minimum	3	Minimum for the primary consolidated cluster
Purchased hosts	5	Maximum of all applicable floors

Resulting utilization

MEASURE	NORMAL	ONE HOST OUT
CPU planning envelope	57.7%	72.1%
Configured memory	61.3%	76.7%

Policy alternative: active-only

Active-only sizing is available when deliberately selected. It limits compute demand to powered-on workload VMs, removes generic CPU and memory headroom, and treats aggregate memory as non-binding. The report must identify that choice because it carries more operational risk than the recommended baseline.

TARGET DESIGN

Node-shape comparison

The recommended node is selected from viable options using total host count and full-silicon VCF licensing. Reduced active cores do not reduce the physical-silicon licensing basis.

SHAPE	HOSTS	ACTIVE	SILICON	VCF	RESULT
E5.192-192	5	192	192	960	Recommended
E4.128-128	6	128	128	768	1 more host
E5.192-96	8	96	192	1,536	No license saving
Standard3.64	11	64	64	704	Invalid: VM fit

VM-fit gate

BM.Standard3.64 is rejected even though its aggregate totals appear attractive: the fictional estate includes a 1.25 TiB VM, which cannot fit on a 1 TiB host. Aggregate capacity never overrides the largest-VM check.

Storage planning

IN USE

127 TiB

inventory observation

PROVISIONED

180 TiB

sizing baseline

PLANNED

225 TiB

+25% headroom

Standard OCVS shapes use OCI Block Volume storage. The capacity figure is a planning input, not a storage design: performance, protection, snapshot growth, datastore layout, and regional limits still require validation.

ASSUMPTIONS AND CHECKS

Before procurement

Validation priorities

- Replace configured-demand proxies with observed CPU, memory, storage-throughput, IOPS, latency, and working-set evidence where available.
- Validate reservations, limits, NUMA topology, latency-sensitive workloads, large pages, passthrough devices, and application-affinity rules.
- The fictional source estate mixes AMD and Intel. Validate migration methods and compatibility for every source subset before selecting one target CPU vendor.
- Confirm target-shape availability, regional capacity, service limits, pricing, VMware bundle, networking, and commercial terms.

Scope and provenance

This is a fully synthetic demonstration. Its inventory and sizing values are invented and are not anonymized or derived from customer data. The report shows the 0.5.0 sizing method, not a purchase recommendation.

Method summary

ELEMENT	METHOD
Compute demand	Configured vCPU and memory from RVTools workload inventory
Resilience	Independent one-host-loss CPU and memory floors
Provider rules	Minimum and maximum cluster size applied separately
Licensing	Full physical silicon cores per purchased host
Selection	Viable node frontier balancing host count and VCF cores

Public methodology references

Oracle Cloud VMware Solution overview and supported shapes

docs.oracle.com/en-us/iaas/Content/VMware/Concepts/ocvsoverview.htm

Broadcom core-counting guidance

knowledge.broadcom.com/external/article/313548/counting-cores-for-vmware-cloud-foundati.html